1 Figure 1.1 shows the variation with the speed v of the force F_D driving a car forward and the resistive force F_R that opposes the motion of the car.

The mass of the car is 1200 kg.

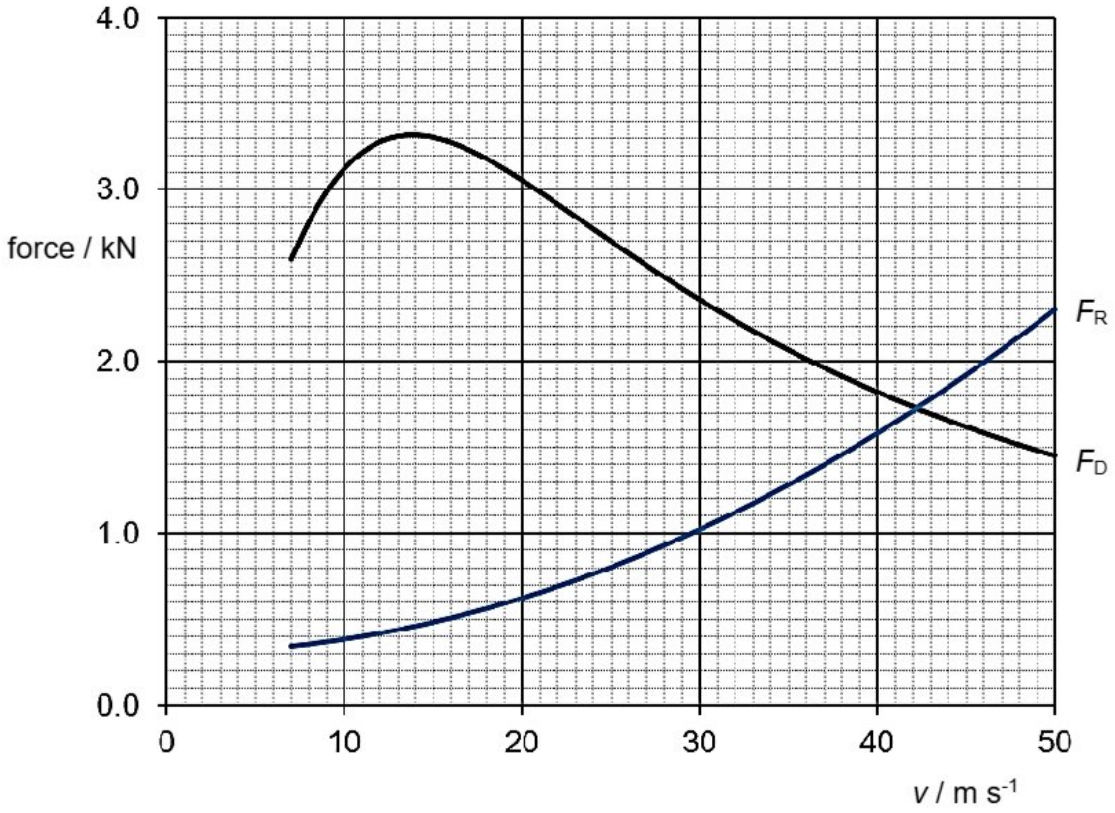


Figure 1.1

(a) Use Figure 1.1 to determine, for the car travelling along a horizontal road at **maximum resultant force,**

(i) the speed at which the car is travelling,

speed = ... m s^{-1} [1]

(ii) its maximum acceleration,

acceleration = ... m s^{-2} [2]

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(iii) the rate of increase of kinetic energy of the car.

rate of increase of kinetic energy = ... W [2]

(b) The car now travels up a slope inclined at an angle to the horizontal.

(i) Explain why, the maximum speed the car can have is less than that when it is travelling on level road.

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..... [2]

(ii) On Figure 1.1, sketch the variation with speed *v* of the total force opposing the motion of the car up the slope.

Label this graph as **S**.

[1]

[Total: 8]